

```
> management <- subset(bank, job == "management" & housing
== "yes" & loan == "yes" & marital == "married" & balance >
5000)

> management
  age      job marital education default balance housing
loan contact day
890  36 management married  tertiary      no   9269    yes
yes cellular  19
1601 32 management married secondary      no   6217    yes
yes cellular  18
1972 32 management married secondary      no   6217    yes
yes unknown  21
4339 50 management married  tertiary      no  19447    yes
yes cellular  21
      month duration campaign pdays previous poutcome y
890   nov      107      2    -1      0 unknown no
1601  nov      486      2   181      2 failure no
1972  may      486      2    -1      0 unknown no
4339  nov      166      1    -1      0 unknown no
```

The `read.csv2()` R function uses semicolon (“;”) as the default value for the separator character.

## JSON

R provides support to handle JSON files. A number of libraries are available, and we will install the *jsonlite* library to read and process JSON files.

```
> install.packages("jsonlite")
Installing package into '/home/shakthi/R/x86_64-pc-linux-
gnu-library/4.1'
(as 'lib' is unspecified)
--- Please select a CRAN mirror for use in this session ---
trying URL 'https://cloud.r-project.org/src/contrib/
jsonlite_1.7.2.tar.gz'
Content type 'application/x-gzip' length 421716 bytes (411
KB)
=====
downloaded 411 KB

* installing *source* package 'jsonlite' ...
** package 'jsonlite' successfully unpacked and MD5 sums
checked
** using staged installation
** libs
...
* DONE (jsonlite)
```

You can now load the library into the R session using:

```
> library("jsonlite")
```

The following JSON file is an example that provides information on colours:

```
[{
  "color": "black",
  "category": "hue",
  "type": "primary",
  "code": {
    "rgba": [255,255,255,1],
    "hex": "#000"
  }
},
{
  "color": "white",
  "category": "value",
  "code": {
    "rgba": [0,0,0,1],
    "hex": "#FFF"
  }
},
{
  "color": "red",
  "category": "hue",
  "type": "primary",
  "code": {
    "rgba": [255,0,0,1],
    "hex": "#FF0"
  }
},
{
  "color": "blue",
  "category": "hue",
  "type": "primary",
  "code": {
    "rgba": [0,0,255,1],
    "hex": "#00F"
  }
},
{
  "color": "yellow",
  "category": "hue",
  "type": "primary",
  "code": {
    "rgba": [255,255,0,1],
    "hex": "#FF0"
  }
},
{
  "color": "green",
  "category": "hue",
  "type": "secondary",
  "code": {
```